

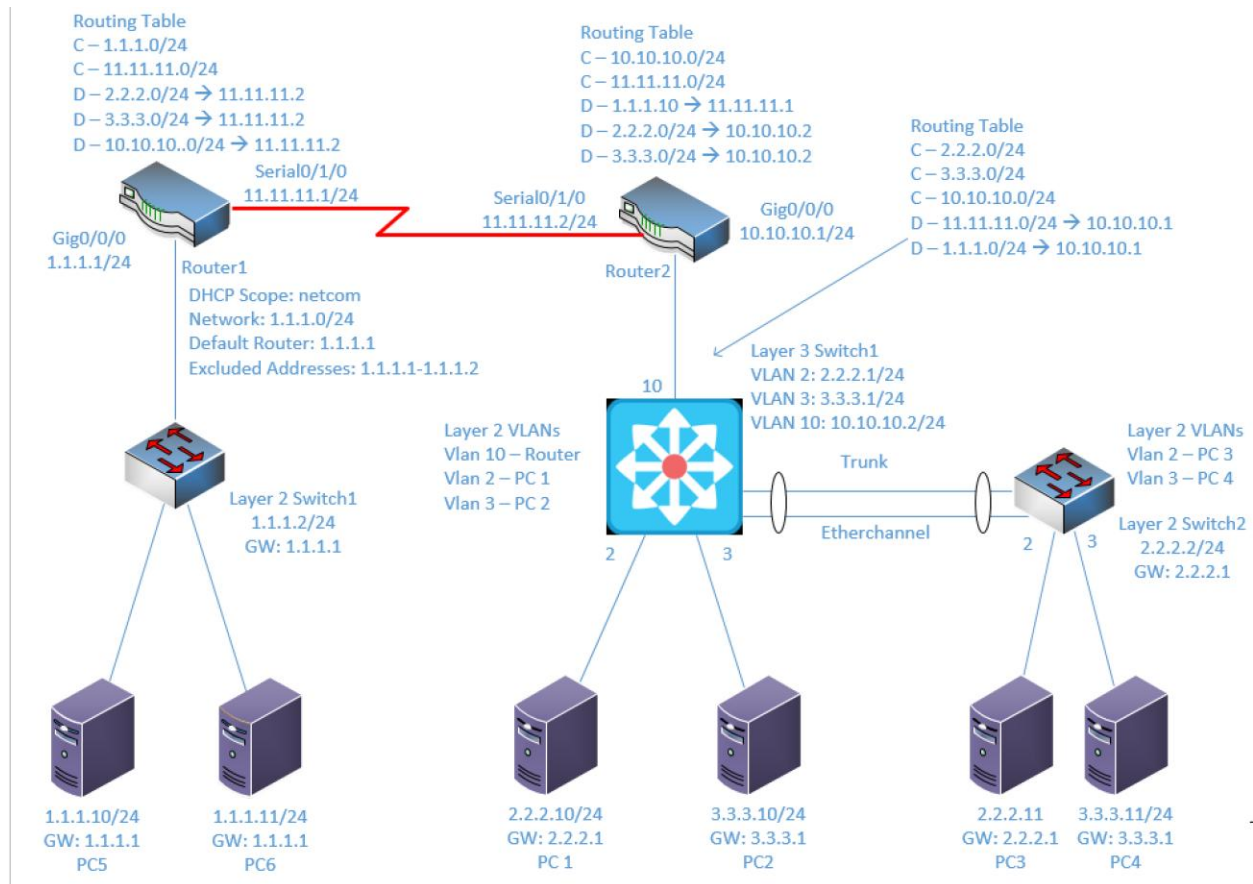
Lab 4

EtherChannel and DHCP

Objective:

The purpose of this lab is for the student to understand, create and implement EtherChannel connections and DHCP. Students will also utilize IOS commands to explore and configure EtherChannel connections and DHCP on network routers and switches.

Diagram:



Components:

- 6 – PC's with an Ethernet NIC
- 1 – 3650 Layer 3 Switch
- 2 – 2960 Layer 2 Switches
- 2 – 4321 Routers (with NIM 2T WAN card)

Useful commands for this lab:

```
show ip route - view routing table
show vlan - View vlan table and port assignments
show ip interface brief - view status of layer 3 ports
show access-lists - view access lists and hit counters
show etherchannel summary - view port channel status
```

Add IP address to Serial Interface (same as Ethernet)

```
interface Serial0/1/0
    ip address 5.5.5.1 255.255.255.0
    no shutdown
```

Enable routing on Layer 3 switch

```
ip routing
```

PPP – Clock rate only on DCE side

```
interface serial0/1/0
    encapsulation ppp
    clock rate 1536000
    ip address 1.1.1.1 255.255.255.0
    no shutdown
```

Assign range of ports to port channel

```
interface range GigabitEthernet1/0/9-10
    channel-group 5 mode on
```

Configure Port Channel for trunking on L3 Switch

```
interface Port-Channel 5
    switchport trunk encapsulation dot1q
    switchport mode trunk
    no shutdown
```

Configure Port Channel for trunking on L2 Switch

```
interface Port-Channel 5
    switchport mode trunk
    no shutdown
```

Configure DHCP on Router

```
ip dhcp pool netcom
  network 1.1.1.0 255.255.255.0
  default-router 1.1.1.1
ip dhcp excluded-address 1.1.1.1 1.1.1.2
```

Configure Port Fast on a port

```
interface GigabitEthernet 1/0/1
  spanning-tree portfast
```

Configure EIGRP

```
router eigrp 5
  network 1.1.1.0 0.0.0.255
  network 11.11.11.0 0.0.0.255
  no auto-summary
```

Procedure:

1. Connect all PC's together and put their static IP addresses, subnet masks and gateways on them. Test connectivity by pinging each machine within the network.
2. Use the diagram above to label the Layer 3 switches VLAN IP addresses and ports below

Vlan	IP Address	Switch Port
2	2.2.2.1/24	2
3	3.3.3.1/24	3
10	10.10.10.2/24	10
Trunk		21-22

3. Use the diagram above to label the routing tables for the Layer 3 switch and routers below.

Router1	Router2	Layer3 Switch
C-1.1.1.0/24	C-10.10.10.0/24	C-2.2.2.0/24
C-11.11.11.0/24	C-11.11.11.0/24	C-3.3.3.0/24
D-2.2.2.0/24-> 11.11.11.2	D-1.1.1.0/24-> 11.11.11.1	C-10.10.10.0/24
D-3.3.3.0/24-> 11.11.11.2	D-2.2.2.0/24-> 10.10.10.2	D-11.11.11.0/24-> 10.10.10.1
D-10.10.10.0/24-> 11.11.11.2	D-3.3.3.0/24-> 10.10.10.2	D-1.1.1.0/24-> 10.10.10.1

4. Place the switch ports into the proper vlans. Configure all of the switches vlan IP addresses.

a. Example

```
interface GigabitEthernet 1/0/1
  switchport mode access
  switchport access vlan 2
  no shutdown
interface vlan 2
  ip address 2.2.2.1 255.255.255.0
  no shutdown
```

5. Enable routing on the Layer 3 switch
6. View the routing table on the Layer 3 switch
7. Verify that the PC's on the switch can ping each other
8. Configure the IP addresses on the fastethernet router interfaces and turn the interfaces on
9. Plug the DCE cable into the router with the 10.10.10.0/24 and 11.11.11.0/24 networks.

10. Plug the DTE cable into the router with the 11.11.11.0/24 and 1.1.1.0/24 networks.
11. Make the serial0/0 interfaces on both routers PPP
12. Make the serial0/0 interface clock rate 56k on the master
13. Place IP addresses on the router serial0/0 interfaces
14. Use a dynamic routing protocol (EIGRP or OSPF) on the two routers and Layer 3 switch. Remember: Dynamic routing protocols require a “network” statement for each interface on the device. Also remember the EIGRP autonomous system number should be the same on all devices.

a. Example: Router 1

```
router eigrp 5
  network 1.1.1.0 0.0.0.255
  network 11.11.11.0 0.0.0.255
  no auto-summary
```

15. View and verify the routing table on both routers and the L3 Switch. Paste below once all 3 L3 devices have all routes.

The image shows two overlapping windows of a Cisco IOS Command Line Interface (CLI). The top window is titled 'Router1' and the bottom window is titled 'Router2'. Both windows have tabs for 'Physical', 'Config', 'CLI' (selected), and 'Attributes'. The CLI text in both windows is identical, showing the following configuration and status information:

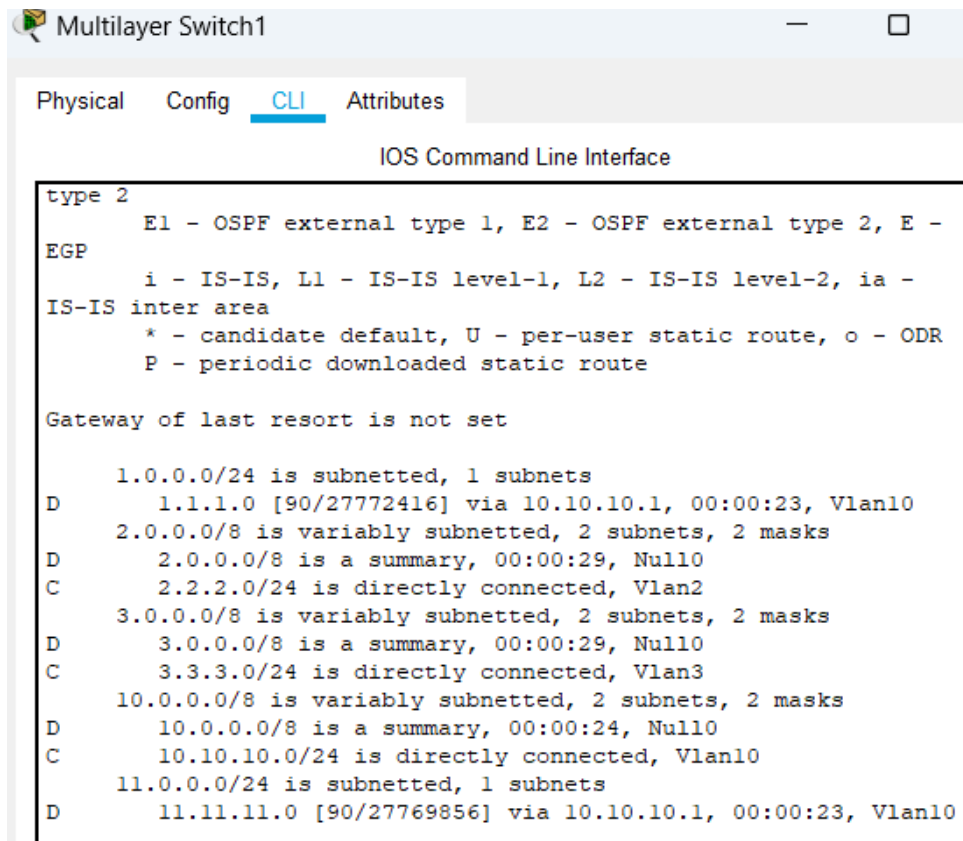
```
IOS Command Line Interface

E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
  i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia -
IS-IS inter area
  * - candidate default, U - per-user static route, o - ODR
  P - periodic downloaded static route

Gateway of last resort is not set

  1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    1.1.1.0/24 is directly connected, GigabitEthernet0/0/0
L    1.1.1.1/32 is directly connected, GigabitEthernet0/0/0
D    2.0.0.0/8 [90/2772416] via 11.11.11.2, 00:01:05,
Serial0/1/0
D    3.0.0.0/8 [90/2772416] via 11.11.11.2, 00:01:05,
Serial0/1/0
  10.0.0.0/24 is subnetted, 1 subnets
D    10.10.10.0/24 [90/2172416] via 11.11.11.2, 00:01:59,
Serial0/1/0
  11.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C    11.11.11.0/24 is directly connected, Serial0/1/0
L    11.11.11.1/32 is directly connected, Serial0/1/0
C    11.11.11.2/32 is directly connected, Serial0/1/0
```

The Router2 window shows the same text, but with a slight difference in the routing table for the 1.1.1.0/24 network, where the interface is GigabitEthernet0/0/0 instead of Serial0/1/0.



The screenshot shows the CLI of a Multilayer Switch1. The 'CLI' tab is selected. The output of the 'show ip route' command is displayed, showing the routing table. The table lists various routes, including OSPF external routes, IS-IS level-1 and level-2 routes, and static routes. The 'Gateway of last resort' is not set. The routes are listed as follows:

```

type 2
    E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
    i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia -
IS-IS inter area
    * - candidate default, U - per-user static route, o - ODR
    P - periodic downloaded static route

Gateway of last resort is not set

    1.0.0.0/24 is subnetted, 1 subnets
D       1.1.1.0 [90/27772416] via 10.10.10.1, 00:00:23, Vlan10
    2.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
D       2.0.0.0/8 is a summary, 00:00:29, Null0
C       2.2.2.0/24 is directly connected, Vlan2
    3.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
D       3.0.0.0/8 is a summary, 00:00:29, Null0
C       3.3.3.0/24 is directly connected, Vlan3
    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
D       10.0.0.0/8 is a summary, 00:00:24, Null0
C       10.10.10.0/24 is directly connected, Vlan10
    11.0.0.0/24 is subnetted, 1 subnets
D       11.11.11.0 [90/27769856] via 10.10.10.1, 00:00:23, Vlan10
  
```

16. Connect 2 interfaces on the Layer 3 switch to 2 interfaces on the Layer 2 switch as the diagram shows
17. On the Layer 3 switch create an EtherChannel with the 2 ports that were just connected to the Layer 2 switch. Use the mode of on and a channel-group number of 1. My example above uses group 5. Make sure to adjust the commands to meet these requirements (channel-group 1 mode on) and (interface port-channel 1)

a. What commands did you enter?

```

Interface range fastethernet0/21-22
Channel-group 1 mode on
  
```

b. What is the name of the new interface that was created?

Interface Port-channel1

18. Select the new interface and place it into trunk mode

a. What commands did you enter?

```
Interface port-channel1
Switchport trunk encapsulation dot1q
Switchport mode trunk
```

19. View the running configuration

- a. Are the physical ports in the port channel also in trunk mode? Yes
- b. Why? The ports belong to the port channel, so their settings are changed when we change settings for the channel.

20. Verify that all 4 PC's on the Layer 2 and Layer 3 switch can ping each other

21. Run a continuous ping from 2.2.2.10 to 2.2.2.11

a. Example

```
ping -t 2.2.2.10
```

The “-t” will send the ping indefinitely or until “Control + C” is pushed or until the Command prompt window is closed.

22. Remove one of the links between the Layer 2 and Layer 3 switch

- a. Did the continuous ping continue or stop? It continued
- b. Why? The connection was not severed; the etherchannel treats both physical ports as one logical port and therefore the “port” did not lose its connection.

23. On the router that has the 1.1.1.0/24 network directly connected, create a dhcp pool with the following settings

- a. Pool Name: NetCom
- b. Network of 1.1.1.0/24
- c. Gateway of 1.1.1.1
- d. Exclude IP address 1.1.1.1

24. What commands did you enter?

```
Ip dhcp pool NetCom
Network 1.1.1.0 255.255.255.0
Default-router 1.1.1.1
Exit
Ip dhcp excluded-address 1.1.1.1
```

25. Disconnect the PC's 1.1.1.10/24 and 1.1.1.11/24 from the switch. Change them to DHCP clients and plug them back in. If the PCs are being stubborn, open a command prompt and enter "ipconfig /renew"

- a. Did they receive IP addresses? No, it says DHCP failed
- b. How long did it take? It looks like a minute or so

26. Disconnect the PC's 1.1.1.10/24 and 1.1.1.11/24 from the switch. Place the switch ports the PC's plug into to be portfast

a. What commands did you enter?

```
Int range fa0/1-2
Sw m a
Spanning-tree portfast
```

- b. Did they receive IP addresses?
Yes
- c. How long did it take?
Literally instantly
- d. Was this faster than before?
Very much so
- e. Why? Because the ports were put into portfast state

27. Can every PC ping each other? Yes!

28. Create the lab diagram below (PCs, router and switch connections) in Packet Tracer or Visio. **Label the IP addresses, subnet masks, and gateways** used by each device. **Label the routing tables next to the switch and router** and the **switches vlan IP addresses and port vlan information**. **Label the port-channel interfaces on the switches**. See the diagram at the beginning of the lab for a suggestion on how to do this.

